



Lab #01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

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Course Name: Cloud DevOps

Section: B

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1. lab01_folder_structure.png

Caption: Shows the required Lab01 folder structure with the screenshots directory created.

```
USER@DESKTOP-AUVEU85 MINGW64 ~/Desktop/DevOps (main)
$ pwd
/c/Users/USER/Desktop/DevOps

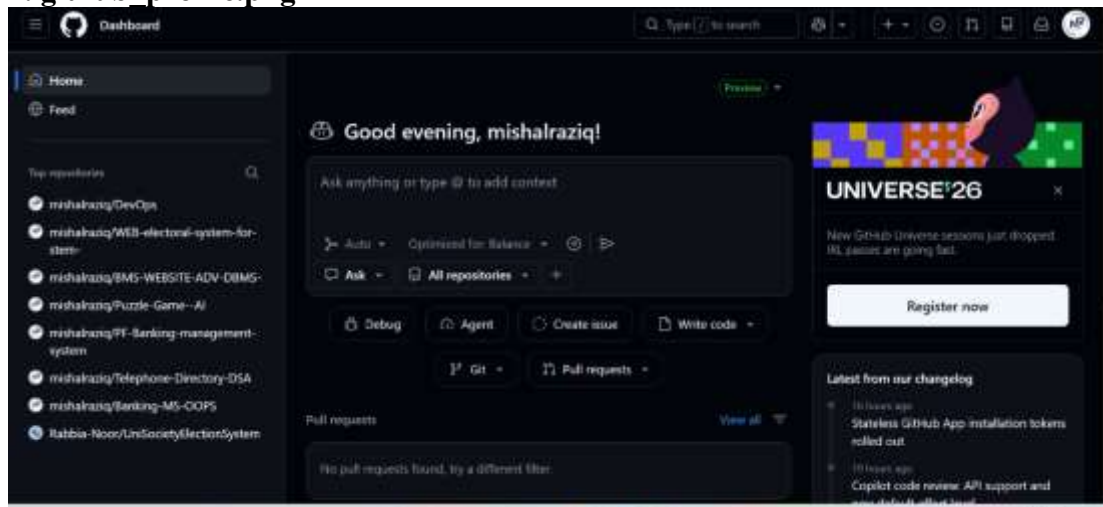
USER@DESKTOP-AUVEU85 MINGW64 ~/Desktop/DevOps (main)
$ ls -la
total 32
drwxr-xr-x 1 USER 197121 0 Oct  3 19:25 ./
drwxr-xr-x 1 USER 197121 0 Oct  3 19:23 ../
drwxr-xr-x 1 USER 197121 0 Oct  3 19:37 .git/
drwxr-xr-x 1 USER 197121 0 Oct  3 19:25 Labs/

USER@DESKTOP-AUVEU85 MINGW64 ~/Desktop/DevOps (main)
$ find Labs -maxdepth 3 -print
Labs
Labs/Lab01
Labs/Lab01/screenshots
```

Caption: Shows the required Lab01 folder structure with the screenshots directory created.

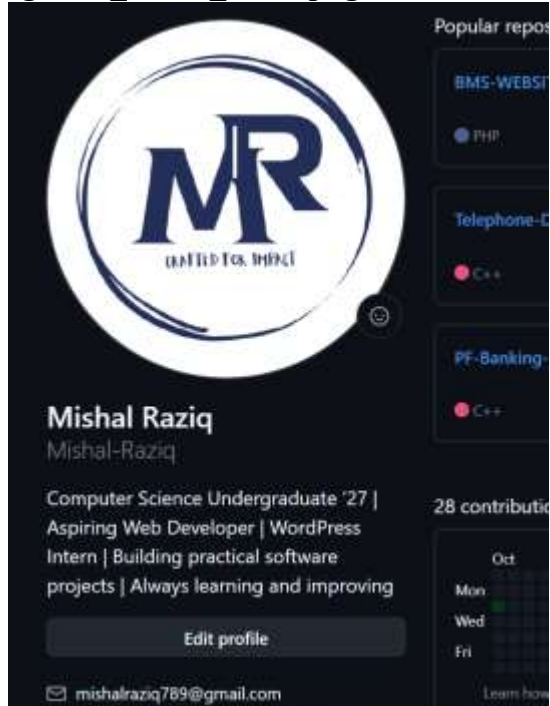
Task 1: Create and Configure Your GitHub Account

2. github_profile.png



Caption: Shows my GitHub profile with my actual name and GitHub username.

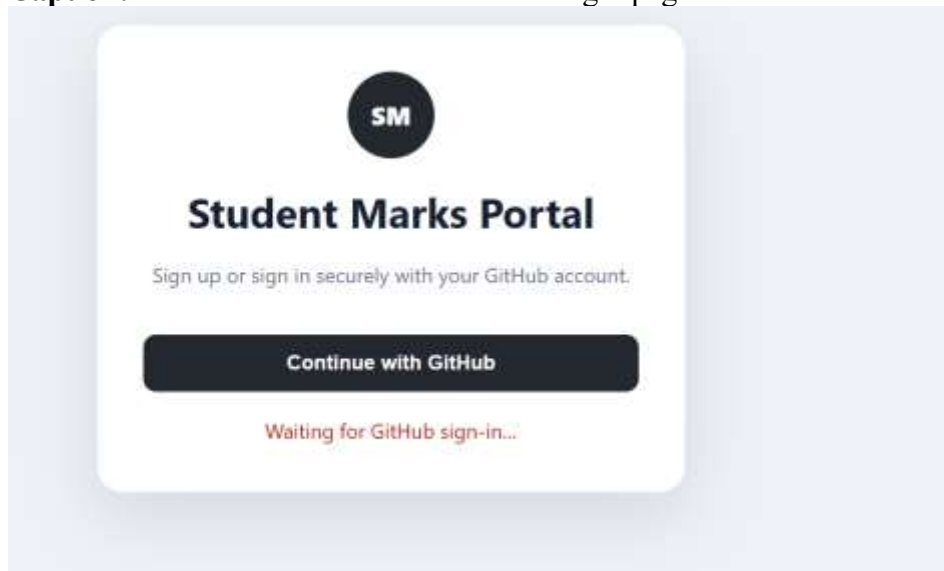
3. github_actual_name.png



Task 2 — Register in the Student Marks Portal

4. portal_login_page.png

Caption: Shows the Student Marks Portal login page.



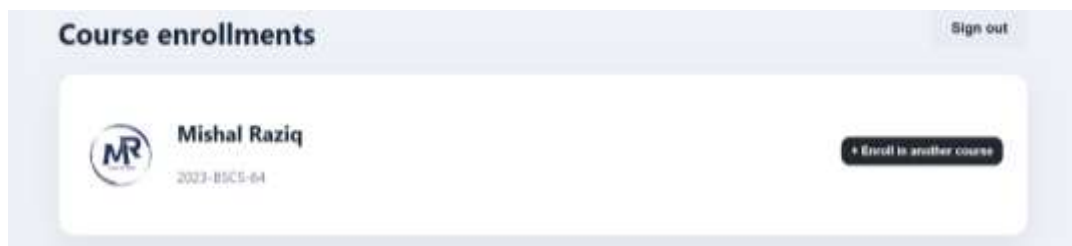
5. portal_github_profile.png

Caption: Shows my GitHub profile information after signing in to the Student Marks Portal.



6. portal_enrollment_submitted.png

Caption: Shows my submitted enrollment information in the Student Marks Portal.



Task 3 — Download VMware Workstation Pro and Ubuntu Server

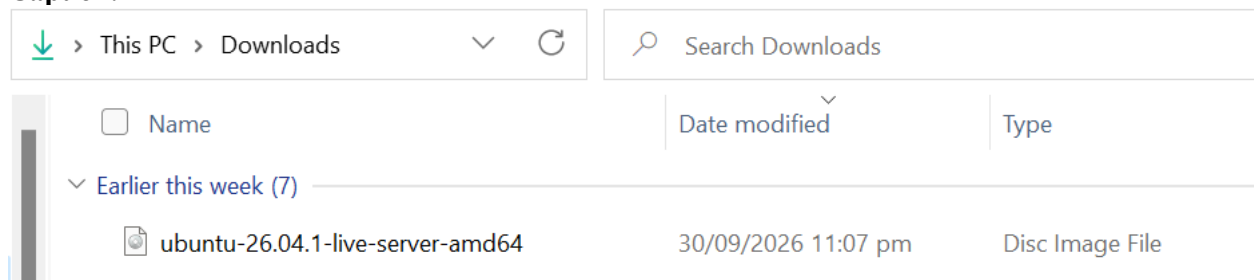
7. vmware_workstation.png

Caption: Shows VMware Workstation Pro running successfully on my Windows system.

 VirtualBox-7.2.4-170995-Win 22/12/2025 1:18 pm Application

8. ubuntu_server_iso.png

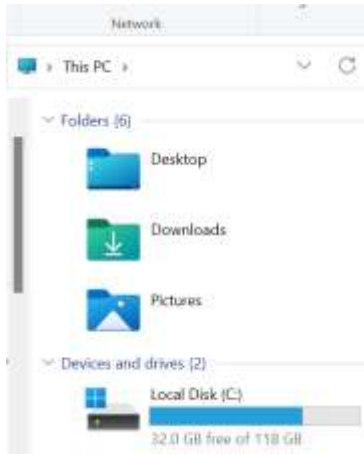
Caption: Shows the downloaded Ubuntu Server LTS ISO file.



9. available_storage.png

Caption: Shows the available storage space required for the Ubuntu Server virtual machine.

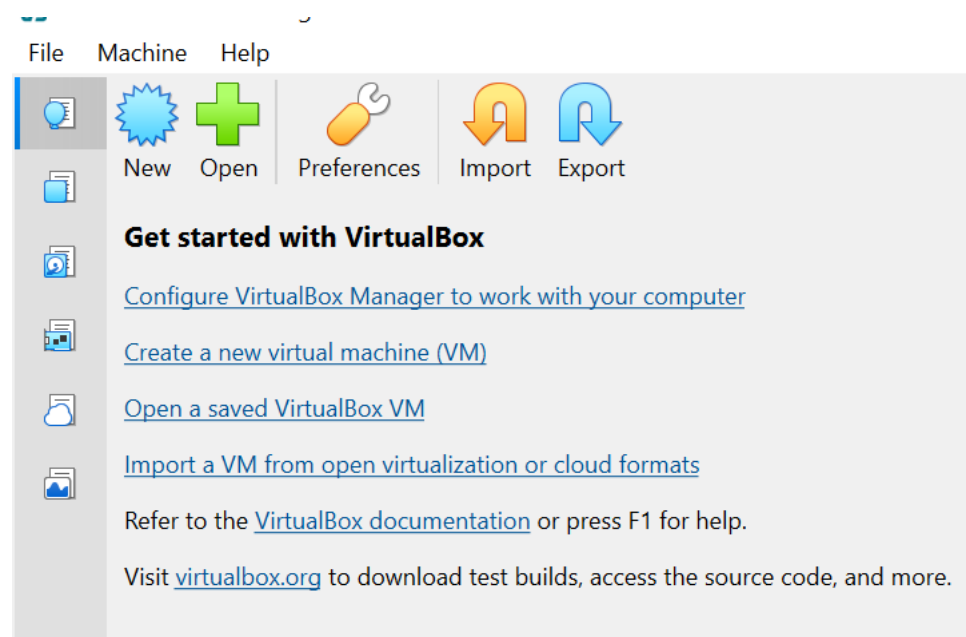
Task 3: Download VMware Workstation Pro and Ubuntu Server



Task 4 — Create the Ubuntu Server Virtual Machine

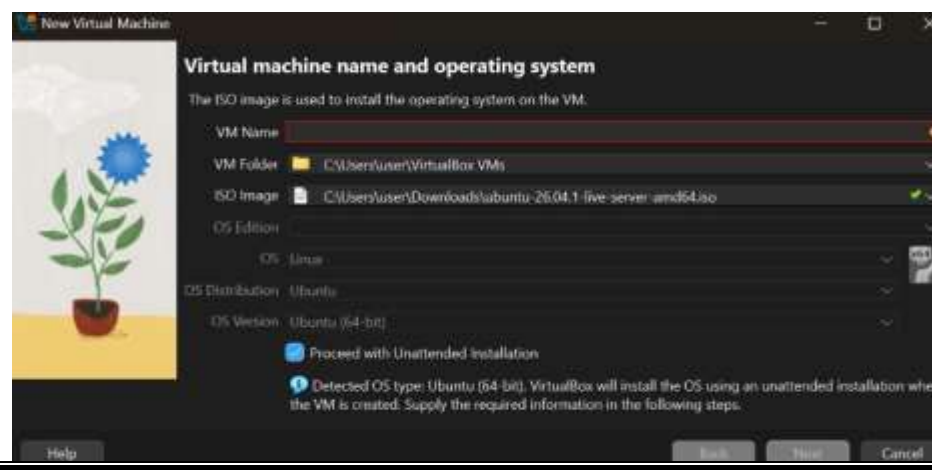
10. vm_creation_wizard.png

Caption: Shows the VMware New Virtual Machine Wizard used to create the Ubuntu Server VM.



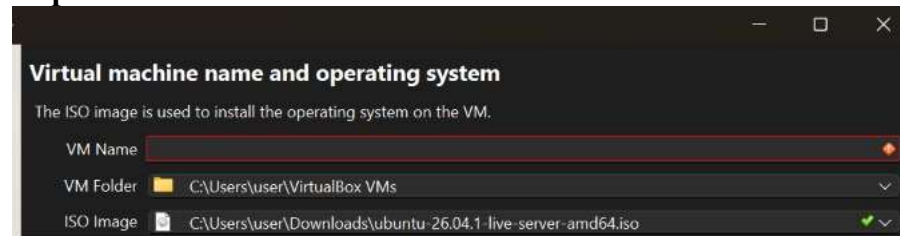
11. vm_typical_configuration.png

Caption: Shows the Typical recommended virtual machine configuration selected.



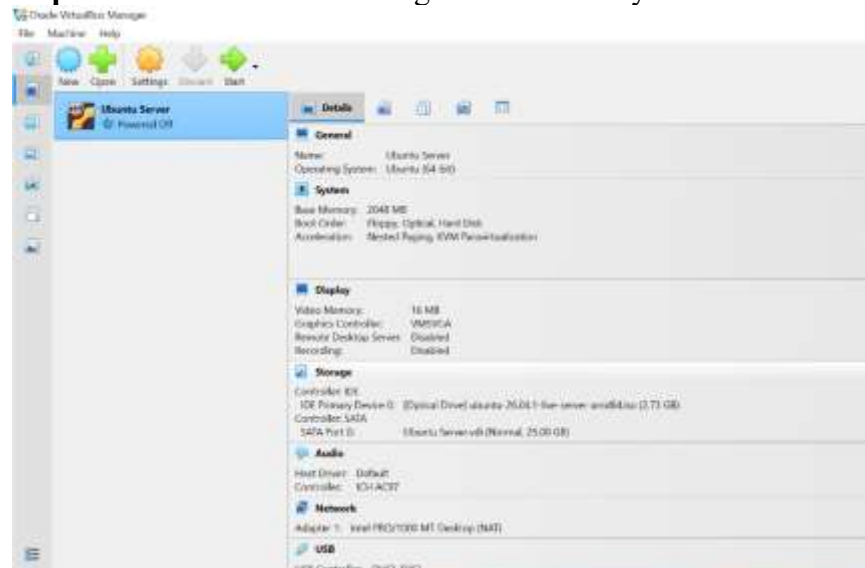
12. vm_ubuntu_iso_selected.png

Caption: Shows the Ubuntu Server ISO selected as the installation media.



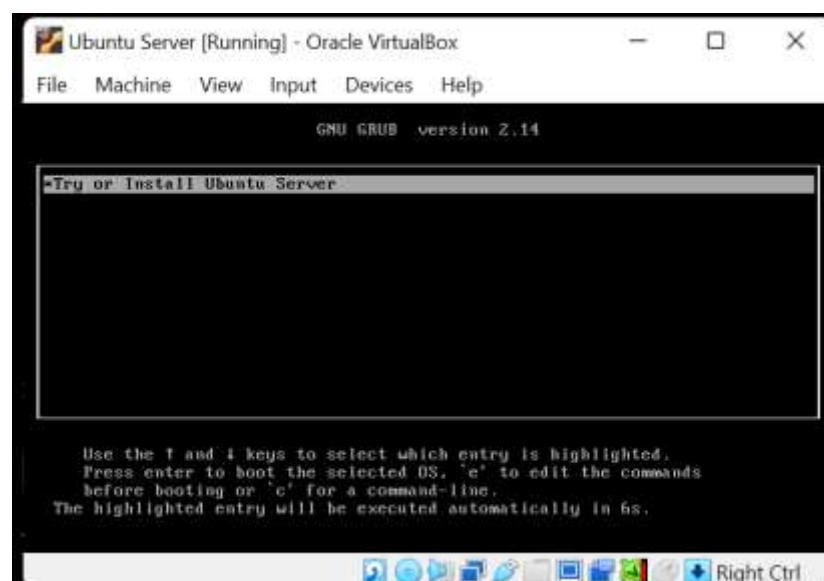
13. vm_configuration_summary.png

Caption: Shows the final configuration summary of the Ubuntu Server virtual machine.



14. ubuntu_installer_boot.png

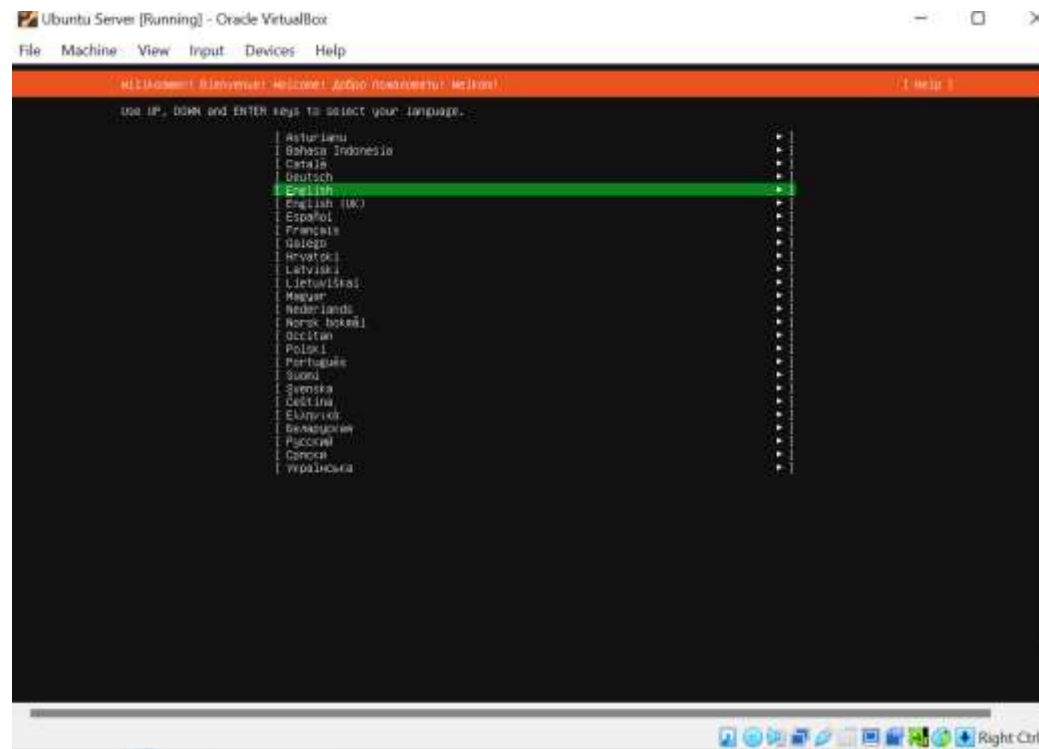
Caption: Shows the Ubuntu Server installer boot screen after starting the virtual machine.



Task 5 — Choose the Installation Language

15. ubuntu_language.png

Caption: Shows the selected language for the Ubuntu Server installation.



Task 6 — Configure the Keyboard Layout

16. ubuntu_keyboard_layout.png

Caption: Shows the selected keyboard layout for the Ubuntu Server installation.



17. ubuntu_keyboard_variant.png

Caption: Shows the selected keyboard variant for the Ubuntu Server installation



Task 7 — Choose the Ubuntu Server Installation Type

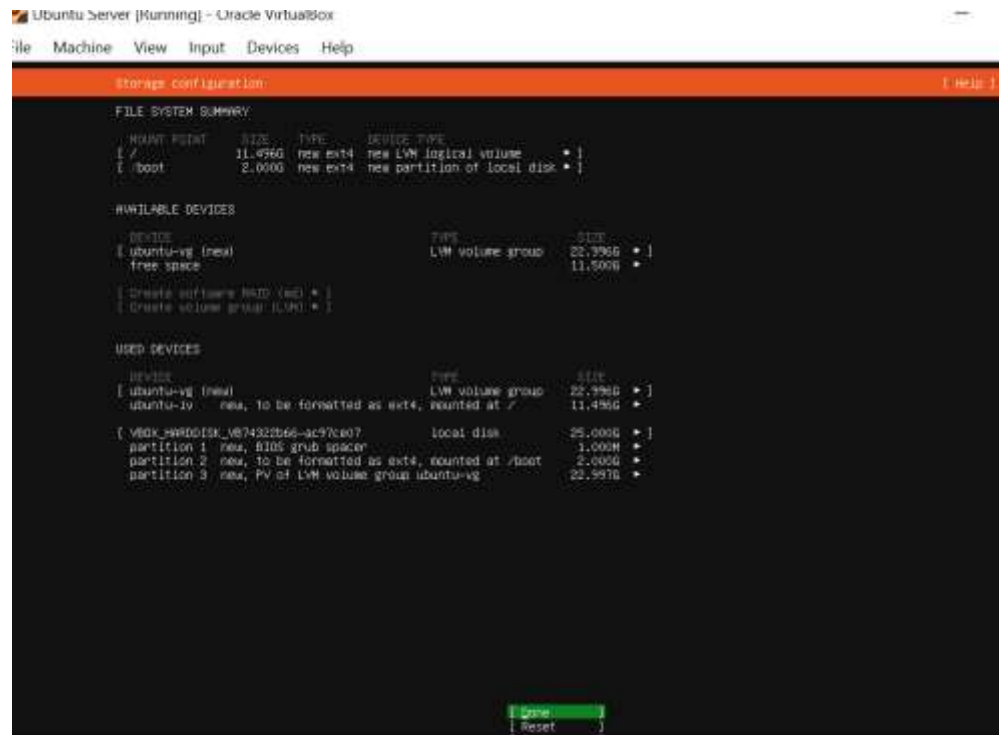
18. ubuntu_installation_type.png

Caption: Shows the normal Ubuntu Server installation option selected.



22. ubuntu_storage_configuration.png

Caption: Shows the proposed storage configuration for the Ubuntu Server virtual machine.



Task 10 — Select the Installation Disk

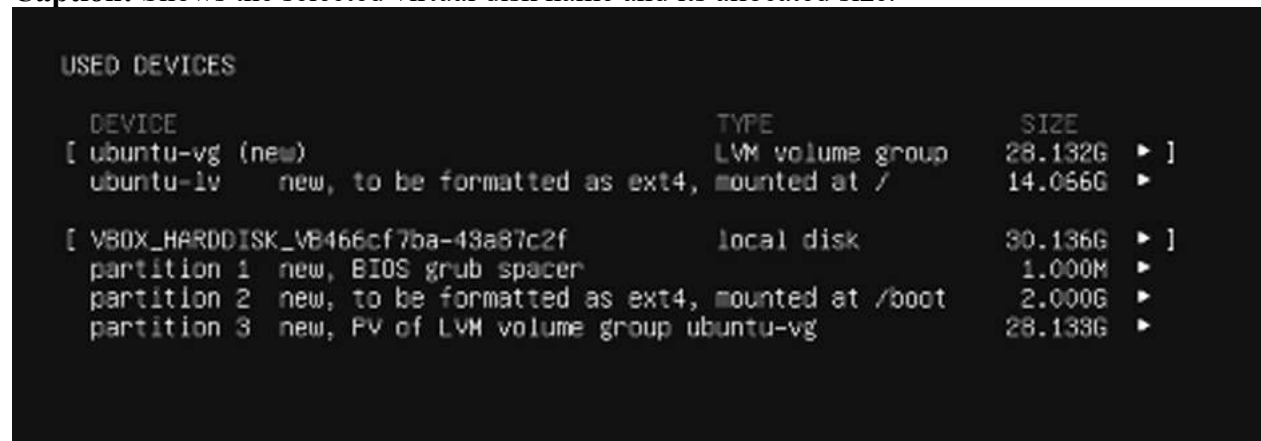
23. ubuntu_selected_disk.png

Caption: Shows the VMware virtual disk selected for the Ubuntu Server installation.



24. ubuntu_disk_details.png

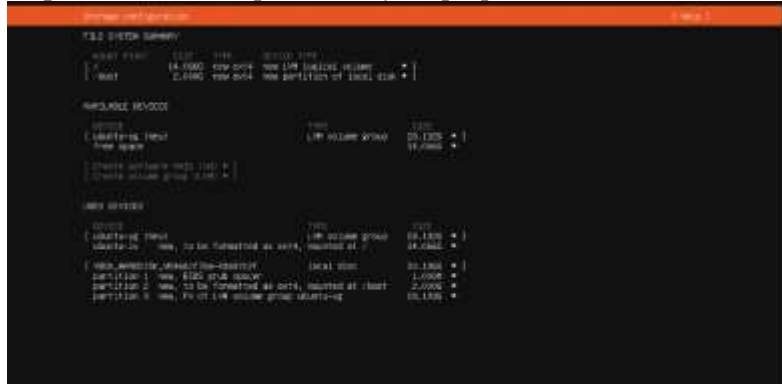
Caption: Shows the selected virtual disk name and its allocated size.



Task 11 — Review the Partitions

25. ubuntu_partition_layout.png

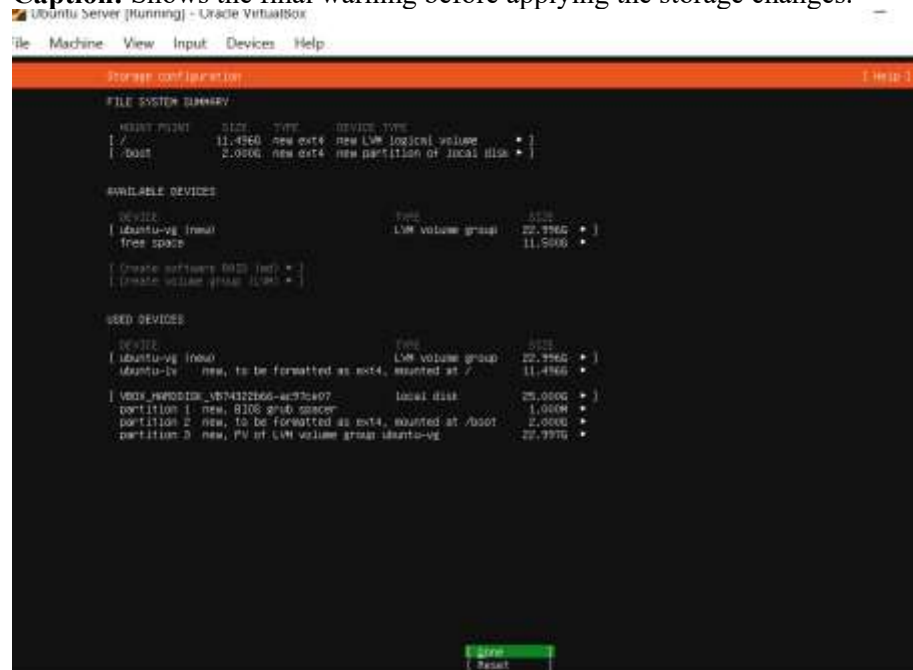
Caption: Shows the partition layout proposed for the selected VMware virtual disk.



Task 12 — Confirm the Storage Changes

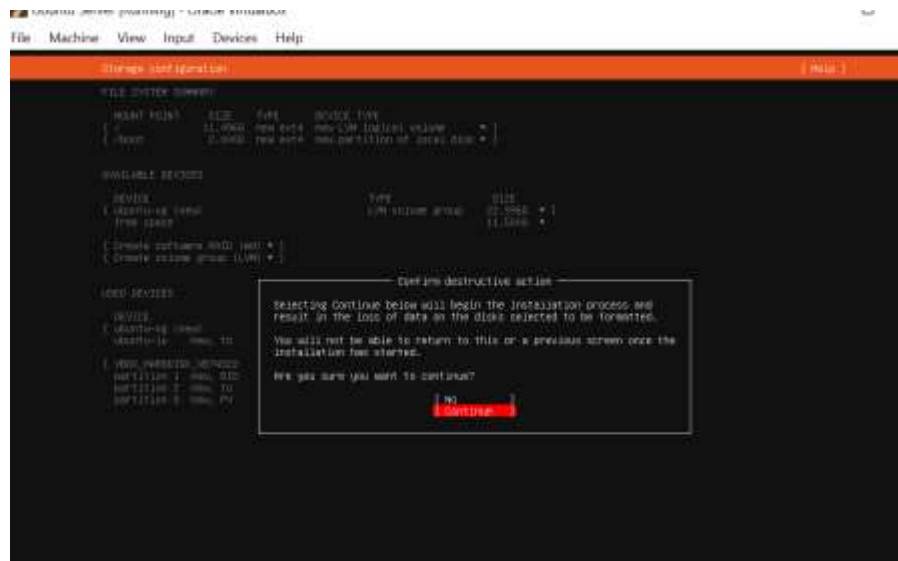
26. ubuntu_storage_warning.png

Caption: Shows the final warning before applying the storage changes.



27. ubuntu_storage_confirmed.png

Caption: Shows the confirmation of the Ubuntu Server storage configuration.



Task 13 — Configure the Ubuntu Profile

28. ubuntu_actual_name.png

Caption: Shows my complete actual name entered in the Ubuntu profile.



29. ubuntu_server_name.png

Caption: Shows the Ubuntu server name configured as ubuntu.



30. ubuntu_username.png

Caption: Shows my GitHub username entered as the Ubuntu username.



31. ubuntu_profile_setup.png

Caption: Shows the completed Ubuntu profile configuration before installation.



Task 14 — Complete the Software Installation

32. ubuntu_installation_progress.png

Caption: Shows the Ubuntu Server software installation in progress.

```
Installing system [Info]

subiquity/RS/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
curtin command install
executing curtin install partitioning step
curtin command install
configuring storage
running 'curtin block-opts simple'
curtin command block-opts
removing previous storage devices
configuring disk: disk-sda
configuring partition: partition-0
configuring partition: partition-1
configuring partition: partition-2
configuring format: format-0
configuring lvm.vg: lvm_vggroup-0
configuring lvm.partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cpi:///tmp/teepr/v011hg/extract
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring sshd service
configuring raid (raid0) service
configuring NTP over TCP
installing kernel

[ View full log ]
```

Task 15 — Reboot and Verify the Ubuntu Identity

33. ubuntu_installation_complete.png

Caption: Shows the Ubuntu Server installation completed successfully.

```
Installation complete [Info]

acquiring and extracting image from cpi:///tmp/teepr/v011hg/extract
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring sshd service
configuring raid (raid0) service
configuring NTP over TCP
installing kernel
setting up dmcc
acquiring networking config
writing etc/hostname
configuring postfix
installing packages on target system
configuring ufw: ufw-agent on target
configuring kernel crash dump settings
final kernel configuration
configuring target system bootloaders
installing and to target device
copying metadata from rootfs
final system configuration
curtin command in-target
curtin command in-target
calculating extra packages to install
installing openssh-server
extracting openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/run-builtins:
subiquity/late/run-builtins: supplied

[ View full log ]
[ Reboot Now ]
```

34. ubuntu_login_screen.png

Caption: Shows the Ubuntu Server login screen after rebooting the virtual machine.

```

ubuntu login:
ubuntu login: mishalraziq
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Oct  3 08:32:13 AM UTC 2026

System load:          0.07
Usage of /:           47.2% of 11.21GB
Memory usage:         14%
Swap usage:           0%
Processes:            102
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:feda:2f60

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

mishalraziq@ubuntu:~$

```

35. ubuntu_terminal_login.png

Caption: Shows the successful terminal login to the installed Ubuntu Server.

```

ubuntu login:
ubuntu login: mishalraziq
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Oct  3 08:32:13 AM UTC 2026

System load:          0.07
Usage of /:           47.2% of 11.21GB
Memory usage:         14%
Swap usage:           0%
Processes:            102
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:feda:2f60

Expanded Security Maintenance for Applications is not enabled.

```

36. ubuntu_identity_verified.png

Caption: Shows the verified Ubuntu terminal prompt using my GitHub username and server name ubuntu.

```

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

mishalraziq@ubuntu:~$

```

Task 16 — Find the Ubuntu Server IP Address

37. ubuntu_ip_addr_command.png

Caption: Shows the ip addr command and its output in the Ubuntu Server terminal.

applicable idm.

```
mishalraziq@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:da:2f:60 brd ff:ff:ff:ff:ff:ff
    altname enx080027da2f60
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85894sec preferred_lft 85894sec
    inet6 fd17:625c:f037:2:a00:27ff:feda:2f60/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86340sec preferred_lft 14340sec
    inet6 fe80::a00:27ff:feda:2f60/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
mishalraziq@ubuntu:~$
```

38. ubuntu_ip_address.png

Caption: Shows the correct IPv4 address assigned to the Ubuntu Server.

```
mishalraziq@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:da:2f:60 brd ff:ff:ff:ff:ff:ff
    altname enx080027da2f60
    inet 192.168.0.105/24 metric 100 brd 192.168.0.255 scope global dynamic enp0s3
        valid_lft 86290sec preferred_lft 86290sec
    inet6 fe80::a00:27ff:feda:2f60/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
mishalraziq@ubuntu:~$
```

Task 17 — Connect to Ubuntu Server from Windows

39. windows_ssh_command.png

Caption: Shows the SSH command used to connect from Windows to the Ubuntu Server.

```
C:\Users\USER>ssh mishalraziq@10.0.2.15
ssh: connect to host 10.0.2.15 port 22: Connection timed out

C:\Users\USER>ssh mishalraziq@192.168.0.105
The authenticity of host '192.168.0.105 (192.168.0.105)' can't be established.
ECDSA key fingerprint is SHA256:/1KiOUjh0rLFH3nNhtLIRTU7bkp2KgVcJdk2QRMTHTA.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

40. windows_ssh_fingerprint.png

Caption: Shows the SSH server fingerprint acceptance during the first remote connection.

```
C:\Users\USER>ssh mishalraziq@192.168.0.105
The authenticity of host '192.168.0.105 (192.168.0.105)' can't be established.
ECDSA key fingerprint is SHA256:/1KiOUjh0rLFH3nNhtLIRTU7bkp2KgVcJdk2QRMTHTA.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

41. windows_ssh_login.png

Caption: Shows the successful SSH login from Windows to the Ubuntu Server.

```
C:\Users\USER>ssh mishalraziq@192.168.0.105
The authenticity of host '192.168.0.105 (192.168.0.105)' can't be established.
ECDSA key fingerprint is SHA256:/1KiOUjh0rLFH3nNhtLIRTU7bkp2KgVcJdk2QRMTHTA.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

42. windows_ssh_identity.png

Caption: Shows the remote SSH terminal prompt confirming my Ubuntu username and server identity.

```
Keywor21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

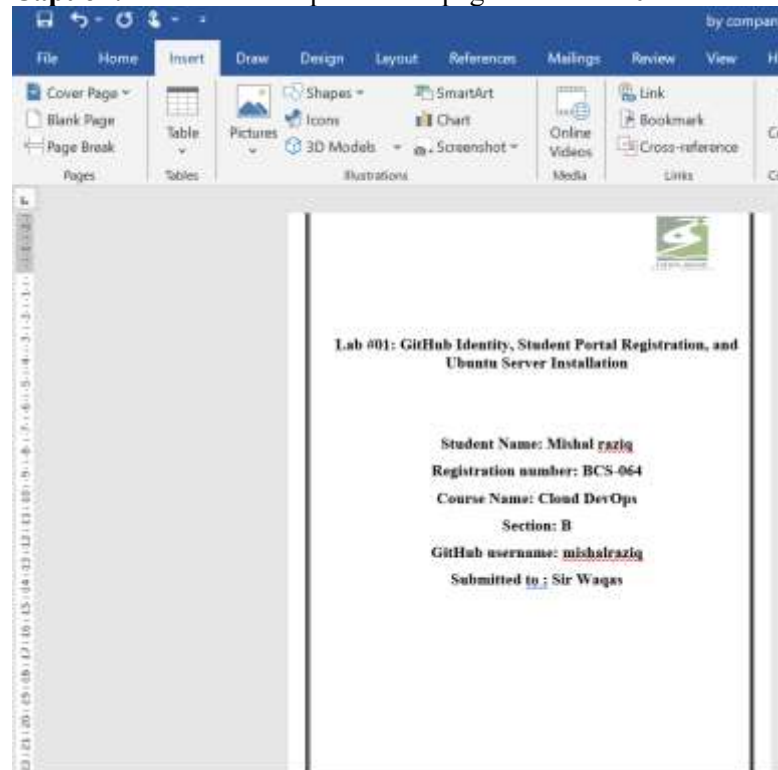
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

❏ C
mishalraziq@ubuntu:~$
```

Task 18 — Prepare Lab1 Solution PDF

43. solution_title_page.png

Caption: Shows the completed title page of the Lab 01 solution document.



44. lab1_solution_pdf.png

Caption: Shows the exported Lab1_Solution.pdf file prepared for submission.

